

MAYA B. MATHUR, PhD

Associate Professor

Departments of Medicine (Biomedical Informatics) & Pediatrics
School of Medicine
Stanford University
mmathur@stanford.edu
[Google Scholar profile](#)

RESEARCH INTERESTS – Statistical Methodology

Evidence synthesis, meta-analysis, causal inference, missing data, graphical models, epidemiologic methods

RESEARCH INTERESTS – Empirical

Humane and sustainable food systems and policy, behavior interventions, experimental social sciences

PRIMARY ACADEMIC POSITIONS

12/2024 – present **Associate Professor**

Primary appointments: Departments of Medicine (Quantitative Sciences Unit in Biomedical Informatics) and Pediatrics

Courtesy appointment: Epidemiology and Population Health
Stanford University

4/2020 – 11/2024 **Assistant Professor**

Primary appointments: Quantitative Sciences Unit, Biomedical Informatics, Pediatrics

Courtesy appointment: Epidemiology and Population Health
Stanford University

10/2019 – 3/2020 **Instructor**

Quantitative Sciences Unit, Biomedical Informatics, Pediatrics
Stanford University

9/2018 – 8/2019 **Postdoctoral fellow**

Department of Epidemiology
Harvard University
Advisor: Dr. Tyler VanderWeele

6/2014 – 9/2019 **Biostatistician**

Quantitative Sciences Unit
Stanford University

6/2013 – 6/2014 **Biostatistician**

Lab of Dr. Nayer Khazeni
Pulmonary & Critical Care Medicine
Stanford University

OTHER ACADEMIC POSITIONS

7/2020 - present **Associate Director**
Stanford Data Sciences' Center for Open and Reproducible Science
(SDS-CORES)
Stanford University

EDUCATION

9/2015 - 8/2018 **PhD Biostatistics**
Harvard University
Advisor: Dr. Tyler VanderWeele
Dissertation: "Statistical methods for evidence synthesis"

11/2011 - 6/2013 **MS Statistics**
Stanford University
(Graduation with Distinction)

9/2009 - 6/2013 **BA Psychology**
Stanford University
(Phi Beta Kappa)

SELECTED HONORS AND AWARDS

(Additional honors for specific papers and proceedings appear under "Publications". Grants appear under "Research Grants and Gifts".)

2026 **Educational Excellence Award**
Computational Medicine Division, Stanford University School of Medicine

2025–2027 **Scholars Research Fellowship**
The Brooks Institute for Animal Law & Policy

2024 **Early Career Epidemiologist Award**
American College of Epidemiology

2022 **Early Career Award**
Society for Research Synthesis Methodology

2022 **McCormick and Gabilan Faculty Award**
Stanford University

2022 **Brian MacMahon Early Career Award**
Society for Epidemiologic Research

2018 **Young Investigator Award**
American Statistical Association, Section on Statistics in Epidemiology

2017 **Distinction in Teaching**
Harvard University

2016 **Editor's Choice paper** (for Mathur & Reichling (2016))
Science Magazine

- 2015 **National Defense Science & Engineering Graduate Fellowship**
- 2013 **Psychology One Research Scholar**
Stanford University
- 2012 **Excellence in Undergraduate Teaching**
Stanford University

TEACHING – Stanford University

- 2023 **Health Research & Policy Department**
Instructor for course “Methods for Reproducible Population Health and Clinical Research”
- 2022 **Department of Pediatrics**
Instructor for independent study on design and conduct of choice-architecture experiments
- 2021–2022 **Biomedical Informatics Program**
Ethics lectures on reproducibility for National Institutes of Health training grant program
- 2021 **Stanford Center for Open and Reproducible Science (CORES)**
Co-instructor for the [Fall Lecture Series](#) of CORES
- 2021–2022 **Department of Pediatrics**
Instructor for independent study on statistical method development
- Summer 2014 **School of Medicine**
Biostatistics mentor for “Intensive Course for Clinical Research” for clinical investigators
- Spring 2012 **Department of Psychology**
TA for “Introduction to Statistical Methods (Precalculus)” for undergraduates
Excellence in Undergraduate Teaching Award

TEACHING – Harvard University

- 2020–present **Population Health Sciences Program**
Guest lecturer on mediation, interaction, replication, and meta-analysis
for PhD-level course: “Quantitative methods for the population health sciences”
- 2017–2019 **Population Health Sciences Program**
TA for PhD-level course: “Quantitative methods in population health sciences”
Distinction in Teaching Award (2017)

TEACHING – Other Venues

- 2021 **Joint Statistical Meetings**
Co-instructor for short course:
“Unmeasured confounding in real-world data analysis: strategies and analytics
for sensitivity analyses”
- 2020 **Society for Epidemiologic Research Meeting**
Co-instructor for 4-hour educational symposium (150 registrants):

“E-values, unmeasured confounding, measurement error, and selection bias”

MENTORING AND ADVISING

2025–present	Yasha Iravantchi (MS Student in Epidemiology, Stanford University) My role: Primary advisor
2024–present	Preeti Panda (Assistant Professor in Emergency Medicine (Pediatrics), Stanford University) My role: Methods advisor on KL2
2020–2022	Yuchen Zhang (Statistics research intern, Stanford University) My role: Mentor for statistics internship
2024–2025	Eric Bridgeford (Postdoctoral fellow in Psychology, Stanford University) My role: Postdoctoral co-advisor
2023–present	Giorgio Ricciardiello Mejia (MS Student in Epidemiology, Stanford University) My role: Primary advisor
2023–present	Anaïs Voski (PhD Student in Environment and Resources) My role: Dissertation committee member
2023–present	Anna Thomas (PhD Student in Computer Science, Stanford University) My role: Primary advisor
2022–present	Stephanie Chao (Assistant Professor in Pediatrics, Stanford University) My role: Methods advisor on KL2
2022–present	Rebecca Dang (Instructor in Pediatrics, Stanford University) My role: Mentor on K proposal
2022–2023	Mika Braginsky (Postdoctoral fellow in Biomedical Informatics, Stanford University) My role: Postdoctoral advisor
2021–present	Yi Li (Statistics research intern, Stanford University) My role: Mentor for statistics internship
2021–2022	Leon Tran King (Statistics research intern, Stanford University) My role: Mentor for statistics internship; independent study instructor
2020–2022	Willi Zhang (Statistics research intern, Stanford University) My role: Mentor for statistics internship
2020–present	Pascal Geldsetzer (Assistant Professor in Medicine, Stanford University) My role: Mentor on funded award from Stanford Diabetes Research Center; methods advisor via Stanford KL2 Mentored Career Development Program
2019–2021	Maximilian Maier (Research intern, Harvard University and Stanford University) My role: Summer internship mentor; senior author on student-led publication

Mentee's current role: PhD student, University of Amsterdam

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|--------------|---|
| 2020–present | Lisa Bruckert (Postdoctoral research fellow in Pediatrics, Stanford)
My role: Post-award mentorship for a project funded by an internal grant for Stanford postdoctoral fellows |
| 2021–2022 | Liz Chin (Biomedical Data Science, Stanford University)
My role: Dissertation committee member |

LEADERSHIP – National Positions

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|-----------|--|
| 2023 | National Institutes of Health
Expert panel member for working group on standardizing the conduct of nonrandomized studies in epidemic settings |
| 2023–2025 | Institute of Education Sciences
Statistical and Research Methodology advisory board member for grant on “Consequences of Selective Reporting Bias in Education Research” |

LEADERSHIP – Intramural Positions

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| 2020–present | Stanford Center for Open and Reproducible Science (CORES)
Associate Director |
| 2023–present | Food Systems Working Group
Working group member advising on Stanford's Climate Action Plan |

PROFESSIONAL SERVICE – Editorial and Peer Review Service

Ad hoc peer reviewer for: *American Journal of Epidemiology* (×7), *Annals of Applied Statistics* (×3), *Annals of Internal Medicine* (×2), *Biological Psychiatry*, *Biostatistics*, *Biometrical Journal*, *Biometrics* (×2), *BMC Medical Research Methodology* (×3), *BMJ* (×2), *Cleaner and Responsible Consumption* (×2), *Epidemiology* (×10), *European Journal of Epidemiology*, *International Journal of Epidemiology* (×3), *International Journal of Human-Computer Studies* (×2), *JAMA*, *JAMA Network Open* (×2), *Journal of the Academy of Nutrition and Dietetics* (×2), *Journal of Internal Medicine*, *Journal of Personality*, *Journal of Psychology & Psychotherapy*, *Journal of Sport & Exercise Psychology*, *Liver International* (×2), *Medical Decision Making*, *Meta-Psychology* (×3), *Molecular Carcinogenesis* (×2), *Nature Human Behaviour* (×2), *Observational Studies*, *Open Forum Infectious Diseases*, *Open Mind: Discoveries in Cognitive Science*, *PLOS One*, *Perspectives on Psychological Science* (×2), *Proceedings of the National Academy of Sciences* (×2), *Psychological Bulletin*, *Psychological Methods* (×3), *Psychological Science*, *Psychoneuroendocrinology* (×3), *Research Synthesis Methods* (×2), *Royal Society Open Science* (×7), *Statistics in Medicine* (×2), and *Systematic Reviews* (×2)

PROFESSIONAL SERVICE – Grant Proposal Reviewing

Study section member or reviewer for: NIH (BMRD study section; reviewed R01 and R21 applications), NIH (National Institute on Drug Abuse special emphasis panel; reviewed P01 applications), Templeton Foundation, ADIA Lab, and PCORI (Methods B review panel)

PROFESSIONAL MEMBERSHIPS

Member of: Society for Research Synthesis Methodology and Society for Epidemiologic Research

ADVISORY BOARDS, AFFILIATIONS, AND CONSULTING

Faculty affiliations within Stanford: Bio-X, Wu Tsai Human Performance Alliance, Maternal & Child Health Research Institute, Stanford Program on Rigor & Reproducibility

Faculty affiliations at other universities: Human Flourishing Program (Harvard University)

Research advisory board memberships: Greener By Default

Consulting partnerships: Agency Fund

RESEARCH GRANTS AND GIFTS

Ongoing and Upcoming Support – As Principal Investigator

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|-------------------|--|
| 5/2026 - 4/2027 | Wellspring Philanthropic Fund
“Statistical methods for a tournament of AI prediction of trial outcomes” |
| 5/2026 - 4/2027 | Players Philanthropy Fund
“Experiments on the small-body problem” |
| 11/2023 - 11/2027 | Coefficient Giving (formerly Open Philanthropy)
General gifts supporting the Stanford Humane & Sustainable Food Lab’s research program |
| 9/2024 - 8/2027 | The Brooks Institute for Animal Law & Policy
“Developing and disseminating a multi-modal plant-based intervention in college dining halls” |
| 9/2021 - 8/2026 | National Institutes of Health
R01: “Statistical methods for modern evidence syntheses with multiple biases” |

Ongoing Support – Other

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|-----------------|--|
| 8/2025 - 8/2026 | Doerr School of Sustainability, Stanford University
“True cost of food: California as a testbed for exploring pathways to reduce meat and animal product consumption”
(PI: Deborah Sivas)
My role: Co-investigator, 0% FTE |
| 4/2022 - 6/2026 | National Institutes of Health
R01: “Building a causal pathway framework to identify interventions to eliminate racial/ethnic disparities in severe maternal morbidity”
(PI: Suzan Carmichael)
My role: Co-I, 8-15% FTE |

Completed Support – As Principal Investigator

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|-----------------|---|
| 6/2024 - 6/2025 | Stanford Plant-Based Diet Initiative |
|-----------------|---|

	“Traditional And Contemporary Meat Options Study (TACOS): a hypothetical discrete-choice pilot study of the effect of the presence of plant-based meats on online menu selections”
6/2023 - 8/2024	Food Systems Research Fund “Effectiveness of plant-based analogs for displacing consumption of animal-based foods”
8/2022 - 7/2024	Stanford University McCormick and Gabilan Faculty Award “Development of statistical methods and software for p-hacking in meta-analyses”
10/2023 - 9/2024	Food Systems Research Fund “Effects of introducing plant-based meats on meat consumption in restaurants: Quasi-experimental analysis”
4/2023 - 4/2024	Food Systems Research Fund “Effect of reducing meat portions on meat consumption and satisfaction: Expanded study in a university cafeteria”
7/2022 - 7/2023	Stanford University School of Medicine Early-career award providing general research support
12/2021 - 7/2023	Food Systems Research Fund “Effect of reducing meat portions on meat consumption and satisfaction”
10/2020 - 12/2022	Food Systems Research Fund “Development and validation of a simple educational intervention to encourage plant-based eating”
9/2019 - 8/2021	Pershing Square Fund for Research on the Foundations of Human Behavior “Psychologically-informed appeals to reduce meat consumption: Meta-analysis, theoretical review, and intervention development”
12/2019 - 8/2020	John E. Fetzer Memorial Trust “Sensitivity analysis for publication bias in meta-analyses: Development of novel statistical methods and results in meta-analyses across disciplines”

Completed Support – Other

9/2017 - 6/2023	National Institutes of Health R01: “Wise social psychological interventions to improve outcomes of behavioral weight control in children with obesity” (PI: Thomas N. Robinson) My role: Faculty biostatistician, 5% FTE (2021-2024)
4/2019 - 3/2024	Centers for Disease Control “Packaging and spreading the Stanford Pediatric Weight Control Program: A family-based, group, behavioral weight control program for children with obesity

	and their families” (PI: Thomas N. Robinson) My role: Faculty biostatistician, 5% FTE (2021-2024)
5/2019 - 2/2024	National Institutes of Health K24: “Patient oriented research in vulnerable populations with skin disease” (PI: Eleni Linos) My role: Biostatistician, 10% FTE (2021 - 2024)
1/2022 - 12/2023	Centre for Effective Altruism “Meta-analysis on substitution effects during efforts to reduce animal product consumption”(PI: Jo Anderson) My role: Co-I, 0% FTE (2022 - 2023)
1/2021 - 12/2022	National Institutes of Health R01: “Theory and methods for mediation and interaction” (PI: Tyler VanderWeele) My role: Biostatistician, 10% FTE (2021 - 2022)
1/2021 - 12/2021	Stanford University Maternal & Child Health Research Institute “Measuring children’s early vocabulary using large scale data from diverse families” (PI: Michael C. Frank) My role: Biostatistician, 2% FTE (2021)
3/2017 - 8/2019	Harvard University Mind, Brain, & Behavior Program Graduate Student Award “Uncanny but not confusing: Multisite study of perceptual category confusion in the Uncanny Valley” My role: Lead investigator
9/2015 - 8/2018	United States Department of Defense National Defense Science & Engineering Graduate Fellowship Full support and stipend for PhD studies
4/2017	Harvard University Mind, Brain, & Behavior Program Graduate Student Award For psychology conference attendance
8/2012 - 9/2013	Stanford University Undergraduate Research Grant “Spatiotemporal dynamics and a prognostic model of healthcare utilization following 2009 (H1N1) pandemic in New York state” My role: Lead investigator
10/2012	Stanford University Undergraduate Conference Grant For medical conference attendance
6/2011 - 8/2011	Stanford University Psych-Summer Research Grant “Using a head-mounted camera to investigate social referencing during naturalistic

word learning”

My role: Undergraduate research assistant

STATISTICAL SOFTWARE DEVELOPED – R Packages

1. R package **multibiasmeta**
Conducts corrections and sensitivity analyses for within-study and/or across-study biases in meta-analyses.
Contributors: Braginsky M & **Mathur MB**.
2. R package **metabias**
Provides common components (classes, methods, documentation) for meta-analysis packages.
Contributors: Braginsky M & **Mathur MB**.
3. R package **truncnormbayes**
Conducts Bayesian estimation of the parameters of a truncated normal distribution.
Contributors: Braginsky M, King LT, & **Mathur MB**.
4. R package **phacking**
Conducts sensitivity analysis for p-hacking in meta-analyses.
Contributors: Braginsky M & **Mathur MB**.
5. R package **regmedint**
Conducts regression-based causal mediation analysis.
Contributors: Yoshida K, Li Y, & **Mathur MB**.
6. R package **PublicationBias**
Conducts sensitivity analyses for publication bias in meta-analyses.
Contributors: **Mathur MB**, Wang R, & VanderWeele TJ.
7. R package **MetaUtility**
Contains functions to estimate the proportion of effects stronger than a threshold of scientific importance, to make various effect size conversions, and to compute and format inference in a meta-analysis.
Contributors: **Mathur MB** & VanderWeele TJ.
8. Stata module **EVALUE**
Conducts sensitivity analyses for unmeasured confounding in observational studies.
Contributors: Linden A, **Mathur MB**, VanderWeele TJ.
9. R package **NRejections**
Computes metrics of outcome-wide evidence strength for studies testing multiple correlated outcomes.
Contributors: **Mathur MB** & VanderWeele TJ.
10. R package **EValue**
Conducts sensitivity analyses for unmeasured confounding or selection bias in observational studies and meta-analyses.
Contributors: **Mathur MB**, Ding P, Smith L, & VanderWeele TJ.
11. R package **Replicate**
Conducts statistical analyses for multisite replication projects.
Contributors: **Mathur MB** & VanderWeele TJ.
12. R package **SimTimeVar**
Simulates a longitudinal dataset with time-varying covariates with user-specified correlation

structures across and within clusters.

Contributors: **Mathur MB**, Kappahn K, Garcia A, Desai M, Montez-Rath M.

STATISTICAL SOFTWARE DEVELOPED – Online Tools

1. Online tool to conduct sensitivity analyses for unmeasured confounding in meta-analyses
[\[Link\]](#)
 Contributors: **Mathur MB**, Ding P, Smith L, VanderWeele TJ, Solymos P, Lee J.
2. Online tool to conduct sensitivity analyses for unmeasured confounding in observational studies
[\[Link\]](#)
 Contributors: **Mathur MB**, VanderWeele TJ, Solymos P.

ORGANIZED CONFERENCE SESSIONS

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|------|---|
| 2021 | Joint Statistical Meetings
Chair: “Recent methodological developments and applications in statistical and machine learning approaches for predictive modeling using competing risk data” |
| 2019 | Joint Statistical Meetings (Denver, CO)
Organizer and chair: “Causal inference with non-traditional designs” |

INVITED PRESENTATIONS – Keynote Talks

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|------|---|
| 2024 | Inaugural Berkeley-Stanford Workshop on Veridical Data Science
(Berkeley, CA)
“Replication puzzles and hidden effect modifiers”
[Video of talk] |
| 2021 | Massachusetts Psychiatric Society Annual Psychopharmacology Update Conference
(Waltham, MA)
“Help for clinicians in understanding unmeasured confounding in observational psychiatric research” |

INVITED PRESENTATIONS – International

Venues without locations listed are those that have international audiences but were held virtually.

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|------|---|
| 2025 | International Statistical Institute (ISI) World Statistics Congress (The Hague, NL)
“A common-cause principle for eliminating selection bias in causal estimands via covariate adjustment” |
| 2025 | Department of Epidemiology, University of Copenhagen (Copenhagen, DK)
“Resurrecting complete-case analysis: A defense” |
| 2023 | Online Causal Inference Seminar
“A common-cause principle for eliminating selection bias in causal estimands via covariate adjustment” |
| 2023 | North American Animal Law Conference & Canadian Animal Law Conference
University of Toronto (Toronto, Canada)
“Nudges to reduce consumption of meat and animal products: The state of the scientific evidence”
[Video of talk] |

- 2023 Formo Food Summit (Berlin, Germany)
“Nudges to reduce animal-based food consumption: Current and future research”
- 2023 Evidence Synthesis and Meta-Analysis in R Conference
Invited panelist for session: “Controlling for publication bias: Challenges & future directions”
- 2023 Département d’Études Cognitives, École Normale Supérieure (Paris, France)
“How robust are meta-analyses to publication bias? Sensitivity analyses and empirical findings”
- 2023 Online Causal Inference Seminar
“Discussion of bounds on selection bias”
[\[Video of talk\]](#)
- 2022 International Day of Women in Statistics and Data Science
“Work-life balance: Naïve perspectives from an early-career researcher”
- 2021 37th Annual Meeting of the International Society for Traumatic Stress Studies
“Applied causal mediation analysis”
- 2021 8th Annual Nordic-Baltic Biometric Conference (Helsinki, Finland)
“Avoiding significance fetishism in meta-analyses: Pitfalls and reporting recommendations”
- 2021 16th Annual Meeting of the Society for Research Synthesis Methodology
“Sensitivity analysis for p-hacking in meta-analyses”
- 2021 Centre for Clinical Epidemiology, Lady Davis Institute, Jewish General Hospital (Montreal, Canada)
“How robust are meta-analyses to publication bias? New statistical methods and empirical findings across meta-analyses”
- 2021 Editorial board meeting of *Psychological Bulletin*
“How robust are meta-analyses to publication bias? New statistical methods and empirical findings across meta-analyses”
- 2020 Joint Statistical Meetings
Invited discussant for session: “Unsupervised learning with latent variables for biobehavioral research”
- 2020 Joint Statistical Meetings
“The E-value: Sensitivity analyses for unmeasured confounding and extensions to meta-analysis”
- 2020 Compassion in World Farming (Godalming, UK)
Research Lunch Series
“Reducing meat consumption by appealing to animal welfare: Meta-analysis and evidence-based recommendations”
- 2019 Launch conference for the multinational study:
“REACH Forgiveness Workbook Intervention Randomized Trial”

- City University of Hong Kong (Hong Kong)
 “Statistical analysis proposal and discussion points”
- 2019 Department of Biostatistics, University of Copenhagen (Copenhagen, DK)
 “The E-value: Practical sensitivity analysis and technical considerations”
- 2019 Danish Epidemiology Society (Copenhagen, DK)
 “Sensitivity analysis for unmeasured confounding in studies and meta-analyses”
- 2018 Joint Statistical Meetings (Vancouver, BC)
 “The E-value: Sensitivity analysis, software, and implementation”

INVITED PRESENTATIONS – Extramural presentations in U.S.A.

- 2026 Society for Epidemiologic Research Annual Mid-Year Meeting
 “Estimating conditional means under missingness not at random with incomplete auxiliary variables”
- 2026 Society for Epidemiologic Research Annual Mid-Year Meeting
 “Beyond publication bias: Assessing p-hacking in meta-analyses”
- 2025 University of California at Los Angeles (Los Angeles, CA)
 Practical Causal Inference Lab Seminar
 “Resurrecting complete-case analysis: A defense”
- 2025 University of California at Davis (Davis, CA)
 Epidemiology Seminar
 “A common-cause principle for eliminating selection bias in causal estimands via covariate adjustment”
- 2025 San Francisco Bay Area Chapter of the American Statistical Association
 (San Francisco, CA)
 “A common-cause principle for eliminating selection bias in causal estimands via covariate adjustment”
- 2024 Harvard University (Boston, MA)
 Design of Experimental and Non-experimental Studies Seminar Series
 Department of Health Care Policy
 “Beyond the funnel plot: New methods to address p-hacking and other biases in meta-analysis”
- 2024 Carnegie Mellon University (Pittsburgh, PA)
 Causal Group Seminar
 “A common-cause principle for eliminating selection bias in causal estimands via covariate adjustment”
- 2024 Joint Statistical Meetings (Portland, OR)
 “A common-cause principle for eliminating selection bias in causal estimands via covariate adjustment”
- 2024 University of North Carolina at Chapel Hill (Chapel Hill, NC)

- Causal Inference Research Group Seminar, Department of Epidemiology
 “A common-cause principle for eliminating selection bias in causal estimands via covariate adjustment”
- 2024 Columbia University (New York, NY)
 Causal Learning Group Seminar, Department of Biostatistics
 “A common-cause principle for eliminating selection bias in causal estimands via covariate adjustment”
- 2024 Northwestern University (Chicago, IL)
 Animal Welfare & Climate Change Symposium, Pritzker School of Law
 Invited panelist for session: “Reducing the demand for inexpensive meat”
- 2024 Harvard University (Boston, MA)
 Animal Law & Policy Workshop Series, Harvard Law School
 “Effect of a default portion-size reduction on meat consumption and diner satisfaction: Controlled experiments in Stanford University dining halls”
- 2023 University of Pittsburgh (Pittsburgh, PA)
 Epidemiology Seminar, Department of Epidemiology
 “A common-cause principle for eliminating selection bias in causal estimands via covariate adjustment”
- 2023 Net-Zero MZ-ECO Sustainability Gala (Palo Alto, CA)
 C&L Sustainability Foundation
 Closing speech: “Toward a plant-based food system: Why and how”
- 2023 Wake Forest University (Winston-Salem, NC)
 Department of Biostatistics and Data Science
 “A common-cause principle for eliminating selection bias in causal estimands via covariate adjustment”
- 2023 Menus of Change University Research Collaborative (MCURC) Annual Meeting (Palo Alto, CA)
 “The SCOOP study: Spoon Capacity Outcomes on Protein Serving Size and Satisfaction”
- 2023 Johns Hopkins University (Baltimore, MD)
 Causal Seminar, Department of Biostatistics
 “A common-cause principle for eliminating selection bias in causal estimands via covariate adjustment”
[\[Video of talk\]](#)
- 2023 Mental Health Research Network, Kaiser Permanente (Seattle, WA)
 “Avoiding significance fetishism in meta-analyses: Pitfalls and reporting recommendations”
- 2023 Society for Epidemiologic Research Conference (Portland, OR)
 “Causal graphs clarify connections between confounding and selection bias”
- 2023 Harvard conference on “Forgiveness: Interdisciplinary Perspectives” (Boston, MA)
 Co-presenter: “International REACH Forgiveness intervention: A randomized trial”

- 2023 Eastern North American Region (ENAR) Spring Meeting (Nashville, TN)
“Avoiding significance fetishism in meta-analyses: Pitfalls and recommendations”
- 2022 BIO-PhRMA (Washington, DC)
Workshop on Advancing Analytical Methodologies for Unmeasured Confounders in Real-world Evidence
“Closing perspectives: Confounding and the credibility of observational studies”
- 2022 BIO-PhRMA (Washington, DC)
Workshop on Advancing Analytical Methodologies for Unmeasured Confounders in Real-world Evidence
“E-values for sensitivity analysis and recent extensions”
- 2022 Harvard University (Boston, MA)
Epidemiology Seminar Series
“How robust are meta-analyses to publication bias? Sensitivity analyses and empirical findings”
[\[Video of talk\]](#)
- 2022 Society for Epidemiologic Research Conference (Chicago, IL)
Plenary talk: “Sensitivity analysis for publication bias and p-hacking in meta-analyses”
- 2022 4th Annual Reducetarian Summit (San Francisco, CA)
“Principles for designing rigorous intervention studies of dietary outcomes”
- 2022 American Educational Research Association (Washington, DC)
Systematic Review & Meta-Analysis Seminar Series
“Sensitivity analysis for uncontrolled confounding in meta-analyses: Applications to meta-analysis on BMI and mortality”
[\[Video of talk\]](#)
- 2021 Boston University (Boston, MA)
Causal Inference Seminar, Departments of Statistics and Biostatistics
“Extending the E-value: Sensitivity analyses for unmeasured confounding in studies and meta-analyses”
- 2021 United States Environmental Protection Agency, Region 6 (Dallas, TX)
“Interventions to reduce consumption of meat and animal products: The state of the scientific evidence”
- 2021 Brigham Young University (Provo, UT)
Neuroscience Seminar
“Addressing unmeasured confounding in observational studies using statistical sensitivity analyses”
- 2020 Yale University (New Haven, CT)
Veterans Aging Cohort Study, Statistics Working Group
“An introduction to marginal structural models”
- 2020 Yale University (New Haven, CT)

- Veterans Aging Cohort Study, Statistics Working Group
“An introduction to causal mediation analysis”
- 2019 Yale University (New Haven, CT)
Veterans Aging Cohort Study, Statistics Working Group
“Sensitivity analyses for unmeasured confounding in studies and meta-analyses”
- 2019 12th Annual FDA/AdvaMed Medical Devices and Diagnostics Statistical Issues Conference (Washington, DC)
“Confounding and methods for sensitivity analysis in observational studies”
- 2019 Northwestern University (Evanston, IL)
Prevention Science and Methodology Group, Virtual Grand Rounds
“Sensitivity analyses for unmeasured confounding in studies and meta-analyses”
[\[Video of talk\]](#)
- 2019 University of Alabama at Birmingham (Birmingham, AL)
Biostatistics Seminar
“Sensitivity analysis for unmeasured confounding in studies and meta-analyses”
- 2017 University of Massachusetts at Amherst epidemiology seminar (Amherst, MA)
“The E-value: Sensitivity analysis, software, and implementation”
- 2009 Centers for Disease Control National Immunization Conference (Dallas, TX)
“Predictors of human papillomavirus vaccination and participation in vaccine decision-making among high school girls”
[\[Audio recording of talk\]](#)
- 2008 American Public Health Association National Meeting (San Diego, CA)
Invited panelist for session: “Inspiring a new generation to address global health”
- 2008 California Medical Association Foundation HPV Vaccine Summit Meeting (Sacramento, CA)
“Predictors of human papillomavirus vaccination and participation in vaccine decision-making among high school girls”
- 2005 NASA Jet Propulsion Laboratory Mars Mission Control Team (Pasadena, CA)
“Exploring the Uncanny Valley: Quantitative test of a theory on emotional responses to humanoid robotic faces”

CONTRIBUTED PRESENTATIONS

- 2025 Society for Epidemiologic Research Conference (Boston, MA)
“Resurrecting complete-case analysis: A defense”
- 2023 Society for Epidemiologic Research Conference (Portland, OR)
“The M-value: Sensitivity analysis for missing data”
- 2023 Atlantic Causal Inference Conference (Austin, TX)
“A common-cause principle for eliminating selection bias in causal estimands via covariate adjustment”

- 2021 Society for the Improvement of Psychological Science
 “Statistical frontiers for selective reporting and publication bias”
 Workshop given with co-presenter, James E. Pustejovsky
- 2021 Animal Advocacy Conference
 “Interventions to reduce meat consumption by appealing to animal welfare:
 Meta-analysis and evidence-based recommendations”
- 2021 Evidence Synthesis and Meta-Analysis in R Conference
 “R package PublicationBias: Simple sensitivity analysis for publication bias”
 [\[Video of talk\]](#)
- 2021 Berkeley Institute for Transparency in the Social Sciences Annual Meetings
 “Estimating publication bias in meta-analyses of peer-reviewed studies:
 A meta-meta-analysis”
- 2019 Berkeley Institute for Transparency in the Social Sciences Annual Meeting (Berkeley, CA)
 “Sensitivity analysis for publication bias in meta-analyses”
 [\[Video of talk\]](#)
- 2019 Harvard University Applied Statistics Workshop (Boston, MA)
 “Sensitivity analysis for publication bias and selective reporting in meta-analyses”
- 2018 Joint Statistical Meetings (Vancouver, BC)
 “Multiple imputation strategies for handling missing data when generalizing
 randomized clinical trial findings through propensity score-based methodologies”
- 2018 Joint Statistical Meetings (Vancouver, BC)
 “Sensitivity analysis for unmeasured confounding in meta-analysis”
- 2017 Berkeley Institute for Transparency in the Social Sciences Annual Meeting (Berkeley, CA)
 “New statistical metrics for multisite replications”
 [\[Video of talk\]](#)

BOOKS

1. Frank MC, Braginsky M, Cachia J, Coles NA, Hardwicke TE, Hawkins RD, **Mathur MB**, Williams R. (2025). Experimentology: An open science approach to experimental methods. *MIT Press*, ISBN 978-0-262-55256-1. Also publicly available: <https://doi.org/10.25936/3JP6-5M50>.

PUBLICATIONS – Methods Papers in Statistics/Quantitative Methods Journals

Underlined: Paper led by my student, mentee, or staff scientist.

1. **Mathur MB**, VanderWeele TJ, Shpitser I (2025). A common-cause principle for eliminating selection bias in causal estimands through covariate adjustment. *The Annals of Statistics*, 53(6): 2303-2328.
2. **Mathur MB** (2025). Meta-analysis with Jeffreys priors: Empirical frequentist properties. *Research Synthesis Methods*, 16(1), 87–122.
3. **Mathur MB** (2024). Sensitivity analysis for the interactive effects of internal bias and publication bias in meta-analyses. *Research Synthesis Methods*, 15(1), 21–43.

4. **Mathur MB** (2024). P-hacking in meta-analyses: A formalization and new meta-analytic methods. *Research Synthesis Methods*, 15(3), 483-499.
5. **Mathur MB** & VanderWeele TJ (2023). New metrics for multiple testing with correlated outcomes. *Frontiers in Applied Mathematics and Statistics*, 9:1151314.
6. **Mathur MB** & VanderWeele TJ (2021). Meta-regression methods to characterize evidence strength using meaningful-effect percentages conditional on study characteristics. *Research Synthesis Methods*, 12(6): 731-749.
7. **Mathur MB** & VanderWeele TJ (2021). Estimating publication bias in meta-analyses: A meta-meta-analysis across disciplines and journal tiers. *Research Synthesis Methods*, 12(2): 176-191.
8. VanderWeele TJ, **Mathur MB**, & Chen Y (2020). Rejoinder: The future of outcome-wide studies. *Statistical Science*, 35(3), 479-483. [Invited commentary]
9. **Mathur MB** & VanderWeele TJ (2020). Sensitivity analysis for publication bias in meta-analyses. *Journal of the Royal Statistical Society: Series C*, 69(5):1091-1119.
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11. **Mathur MB** & VanderWeele TJ (2020). New statistical metrics for multisite replication projects. *Journal of the Royal Statistical Society: Series A*, 183(3):1145-1166.
12. Ling A, Montez-Rath MR, **Mathur MB**, Kapphahn K, Desai M. (2020). How to apply multiple imputation in propensity score matching with partially observed confounders: A simulation study and practical recommendations. *Journal of Modern Applied Statistical Methods*, 19(1): eP3439.
13. **Mathur MB** & VanderWeele TJ. (2019). Discussion on the meeting on “Signs and sizes: understanding and replicating statistical findings”. *Journal of the Royal Statistical Society: Series A*, 183(5), 449-469. [Commentary]
14. VanderWeele TJ, Ding P, **Mathur MB** (2019). Technical considerations in the use of the E-value. *Journal of Causal Inference*, 7(2).
15. Desai M, Montez-Rath M, Kapphahn K, Joyce V, **Mathur MB**, Garcia A, Purington N, Owens DK (2019). Missing data strategies for time-varying confounders in comparative effectiveness studies of non-missing time-varying exposures and right-censored outcomes. *Statistics in Medicine*, 38(17), 3204-3220.
16. **Mathur MB** & VanderWeele TJ (2020). Sensitivity analysis for unmeasured confounding in meta-analyses. *Journal of the American Statistical Association*, 115(529), 163-170.
17. **Mathur MB** & VanderWeele TJ (2019). New metrics for meta-analyses of heterogeneous effects. *Statistics in Medicine*, 38(8), 1336-1342.
18. Montez-Rath ME, Kapphahn K, **Mathur MB**, Mitani AA, Hendry DJ, Desai M (2017). Guidelines for generating right-censored outcomes from a Cox model extended to accommodate time-varying covariates. *Journal of Modern Applied Statistical Methods*, 16(1), 86-106.

PUBLICATIONS – Methods Papers in Other Journals

19. Bridgeford EW, Caffo BS, **Mathur MB**, Poldrack RA (in press). How causal perspectives can inform neuroscience data analysis. *Nature Neuroscience*.
20. **Mathur MB**, Shpitser I, VanderWeele TJ (2026). Resurrecting complete-case analysis: A defense. *American Journal of Epidemiology*, 195(6): 1765-1778.

21. Madley-Dowd P, Hughes RA, **Mathur MB**, Heron J, Tilling K (2026). Using directed acyclic graphs to determine whether multiple imputation or subsample multiple imputation estimates of an exposure-outcome association are unbiased. *American Journal of Epidemiology*, 195(2):505-514.
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23. **Mathur MB** & Shpitser I (2025). Pitfalls of imputing using incomplete auxiliary variables. *American Journal of Epidemiology*, 194(6), 1801-1802.
24. **Mathur MB** & Shpitser I (2025). Simple graphical rules for assessing selection bias in general-population and selected-sample treatment effects. *American Journal of Epidemiology*, 194(1), 267–277.
25. **Mathur MB** & Shpitser I (2025). Mathur and Shpitser respond to “The evolution of selection bias in the recent epidemiologic literature – a selective overview”. *American Journal of Epidemiology*, 194(1), 585-586. [Commentary]
26. **Mathur MB** (2025). On the statistical analysis of studies with attention checks. *Advances in Methods and Practices in Psychological Science*, 8(2), 1–16.
27. **Mathur MB** (2025). On the statistical analysis of experiments with manipulation checks. *Advances in Methods and Practices in Psychological Science*, 8(1), 1–11.
28. **Mathur MB** (2024). Assessing robustness to worst-case publication bias using a simple subset meta-analysis. *BMJ*, 384:e076851.
29. **Mathur MB** & Mathur VS (2024). Toward evidence-based communication on overweight body mass index and mortality. *BMC Medicine*, 22(1):183.
30. Li Y, **Mathur MB**, Solomon DH, Glynn RJ, Yoshida K (2023). Effect measure modification by covariates in mediation: Extending regression-based causal mediation analysis. *Epidemiology*, 34(5):661-672.
31. **Mathur MB** (2023). The M-value: A simple sensitivity analysis for bias due to missing data in treatment effect estimates. *American Journal of Epidemiology*, 192(4), 612-620.
32. Li Y, Kaufman J, **Mathur MB** (2023). A brief primer on conducting regression-based causal mediation analysis. *Psychological Trauma: Theory, Research, Practice, and Policy*, 15(6):930-938.
33. **Mathur MB**, Covington C, VanderWeele TJ (2023). Variation across analysts in statistical significance, yet consistently small effect sizes. *Proceedings of the National Academy of Sciences*, 120(3):e2218957120. [Commentary]
34. Fox MP, **Mathur MB**, Matthay EC (2022). Quantifying the impact of bias to inform quality assessments in systematic reviews: The case of perchloroethylene and non-Hodgkin’s lymphoma. *Global Epidemiology*, 4, 100090.
35. **Mathur MB**, Smith LH, Yoshida K, Ding P, VanderWeele TJ (2022). E-values for effect heterogeneity and approximations for causal interaction. *International Journal of Epidemiology*, 51(4), 1268–1275.
36. **Mathur MB** & VanderWeele TJ (2022). Methods to address confounding and other biases in meta-analyses: Review and recommendations. *Annual Review of Public Health*, 43:19–35.
37. Maier M, VanderWeele TJ, **Mathur MB** (2022). Using selection models to assess sensitivity to publication bias: A tutorial and call for more routine use. *Campbell Systematic Reviews*, 2022;18:e1256.

38. **Mathur MB** & VanderWeele TJ (2022). How to report E-values for meta-analyses: Recommended improvements and additions to the new GRADE approach. *Environment International*, 160, 107032. [Commentary]
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40. VanderWeele TJ, Martin J, **Mathur MB** (2020). E-values and incidence density sampling. *Epidemiology*, 31(6), e51-52.
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43. Zhang X, Stamey JD, **Mathur MB** (2020). Assessing the impact of unmeasured confounders for credible and reliable real-world evidence. *Pharmacoepidemiology and Drug Safety*, 29:1219–1227.
44. VanderWeele TJ & **Mathur MB** (2020). Commentary: Developing best-practice guidelines for the reporting of the E-value. *International Journal of Epidemiology*, 49(5), 1495-1497. [Commentary]
45. **Mathur MB** & VanderWeele TJ (2020). Robust metrics and sensitivity analyses for meta-analyses of heterogeneous effects. *Epidemiology*, 31(3):356-358.
46. **Mathur MB** & VanderWeele TJ (2020). A simple, interpretable conversion from Pearson’s correlation to Cohen’s *d* for continuous exposures. *Epidemiology*, 31(2):e16-e18.
47. Linden A, **Mathur MB**, & VanderWeele TJ (2020). Conducting sensitivity analysis for unmeasured confounding in observational studies using E-values: The evalua package. *The Stata Journal*, 20(1): 162-175.
48. **Mathur MB** & VanderWeele TJ (2020). Evidence relating health care provider burnout and quality of care. *Annals of Internal Medicine*, 172(6):e16-e18. [Commentary]
49. VanderWeele TJ & **Mathur MB** (2019). Some desirable properties of the Bonferroni correction: Is the Bonferroni correction really so bad? *American Journal of Epidemiology*, 188(3), 617-618.
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PUBLICATIONS – Empirical and Substantive Papers

55. [Panda P](#), Deng Y, Lynge A, Forest AS, Triolo B, Fang A, Ward V, Wang E, Newberry J, **Mathur MB**, Patel AI (in press). Association between Childhood Opportunity Index and pediatric emergency department utilization of youth with experiences of trafficking: A national multicenter retrospective cohort study. *Pediatric Emergency Care*.
56. Pathmarajah P, Li S, **Mathur MB**, Johnson A, Marinkovich MP, Admani S, Eid E, Tu J, Mitchell DC, Perrone L, Villanueva Gaona R, Teng J, Tang JY, Chiou AS (in press). Validation of the Investigator Global Assessment Scale for epidermolysis bullosa simplex. *JEADV Clinical Practice*.
57. Tsui A, Carstensen A, Kachergis G, Cao A, Abubakar A, Asnake M, Barry O, et al., including **Mathur MB** (2026). Exploring variation in infants' preference for infant-directed speech: Evidence from a multi-site study in Africa. *Developmental Science*, 29(5): e70242.
58. Szabo YZ, Nishimi K, Smirnova M, Niles A, Lerche AS, Tsai H, Afroz S, Huey N, **Mathur MB**, & O'Donovan A (2026). Sex-specific associations of depression with inflammatory markers: A systematic review and meta-analysis. *Translational Psychiatry*, 16(1): 394.
59. [Hope J](#), [Green SA](#), [Peacock JR](#), **Mathur MB** (2026). Effects of adding plant-based menu options on meat selection frequency: A randomized controlled experiment. *Food Quality and Preference*, 143, 105931.
60. [Thomas AT](#), [Yee A](#), **Mathur MB**, [Jurafsky D](#), [Gligorić K](#) (2025). What can large language models do for sustainable food? *Proceedings of the 42nd International Conference on Machine Learning, PMLR* 267:59377-59433.
Note: In computer science, conference proceedings are the primary publication outlet.
61. [Koh HK](#), [Frederick DE](#), [Balboni TA](#), [O'Reilly SM](#), [Kelly JF](#), [Humphreys K](#), [Botticelli M](#), **Mathur MB**, [Psimopoulos C](#), [Long KN](#), [VanderWeele TJ](#) (2026). Spirituality & harmful or hazardous drug use: Meta-analysis of longitudinal studies and implications. *JAMA Psychiatry*, 83(4):364–378.
62. [Tsai JL](#), [Chen DS](#), [Yang AM](#), [Cachia JYA](#), [Blevins E](#), [Ko M](#), **Mathur MB**, et al. (2025). A meta-analytic review of cultural variation in affect valuation. *Psychological Bulletin*, 151(12):1486–1524.
63. [Thomas AT](#), [Hope J](#), **Mathur MB** (2025). Documentary films can increase public interest in plant-based diets in the USA. *Nature Food*, 6(9), 837–842.
64. [Voski A](#), [Braginsky M](#), [Zhang A](#), [Bertoldo J](#), [Egan S](#), [Levig LA](#), [Ihrig MM](#), **Mathur MB** (2025). Effect of a portion-size default nudge on meat consumption and diner satisfaction: Controlled experiments in Stanford University dining halls. *BMC Public Health*, 25 (1), 1434.
65. [Green SA](#), [Smith BA](#), **Mathur MB** (2026). Meaningfully reducing consumption of meat and animal products is an unsolved problem: A meta-analysis. *Appetite*, 216(1), 108233.

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66. [Panda P](#), Deng Y, Fang A, Ward V, Wang E, Newberry J, **Mathur MB**, Patel A (2025). Increased emergency department identification of youth experiencing trafficking during the COVID-19 pandemic: An interrupted time series analysis. *Public Health Reports*, 140(1_suppl), 74S–81S.
67. [Ricotta EE](#), [Carrillo FAB](#), [Angelli-Nichols S](#), [Barugahare J](#), [Benton A](#), [Carlson CJ](#), [Chang-Rabley E](#), [Dean N](#), [Duda SN](#), [Federer L](#), [Fill MMA](#), [LeRoy EC](#), [Linton NM](#), [Lipsitch M](#), **Mathur MB**, [Phelan AL](#), [Rid A](#), [Rosen JB](#), [Sauer L](#), [Sullivan SG](#), [Zendt M](#), [Evans NG](#). (2025). Observational research in epidemic settings: A roadmap to reform. *BMJ Global Health*, 10, e017981.

68. Zhang W, Tynelius P, **Mathur MB**, Quartagno M, Brandén G, Liljeros F, Kosidou K (2025). Temporal trends in sexual identity and sociodemographic disparities in Stockholm, Sweden, 2010–2021. *American Journal of Public Health*, 115(12): 2062-2071.
69. Zhang W, Tynelius P, **Mathur MB**, Quartagno M, Brandén G, Liljeros F, Kosidou K (2024). Population trends and individual fluidity of sexual identity in Stockholm County, 2010 to 2021. *JAMA Network Open*, 7(12):e2447627.
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